

Sequences and limits of sequences



Outline

Definition of sequence and properties

Convergent sequences and definition of the limit of a sequence

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Monotone sequences

Limits of geometric progressions

Definition of sequence and properties



Introduction: Definition of sequence and properties

- ▶ In this section we introduce **sequences**: functions that assign a real number a_n to every natural number n
- ▶ We will see how to represent them graphically, what it means for them to be **bounded**, and what it means that a property holds **eventually**
- ▶ We then define the central concept of **limit**: **convergent**, **divergent** and **irregular** sequences
- ▶ The limit of a sequence is the first step towards calculus: all later limits are built on these ideas

- ▶ Consider the set $\mathbb{N}$ of non-negative integers, ordered according to the natural order

$$\mathbb{N} : 0, 1, 2, 3, \dots, n, \dots$$

Definition 1: Sequence

*A **sequence** is a relation that associates with each natural number $n \in \mathbb{N}$ (or from a certain natural number n_0 onward) a real number $a_n \in \mathbb{R}$.*

- A sequence is therefore a **function**:

$$f : \mathbb{N} \rightarrow \mathbb{R}$$

$$f : n \mapsto a_n$$

or possibly

$$f : \{n \in \mathbb{N} : n \geq n_0\} \rightarrow \mathbb{R}$$

for some fixed integer $n_0 \in \mathbb{N}$.

Definition of sequence and properties

The sequence associates with the **input variable** n the **output value** a_n .

- ▶ The fact that the domain of the function f is the set of natural numbers makes it possible to write the sequence by **listing its values**, in the order in which they follow one another as n increases:

$$a_0, a_1, a_2, \dots, a_n, \dots$$

- ▶ The dots after a_n indicate that we are not considering only the first n terms of the sequence (i.e., a **finite set** of numbers), but the whole sequence of *infinitely many terms* (i.e., an **infinite set** of numbers).

Example: Sequences

$$n \in \mathbb{N} \quad n \mapsto n^2 \quad 0, 1, 4, 9, 16, \dots$$

$$n \in \mathbb{N} \quad n \mapsto (-1)^n \quad 1, -1, 1, -1, 1, \dots$$

$$n \in \mathbb{N} \quad n \mapsto a \in \mathbb{R} \quad a, a, a, a, a, \dots$$

The last sequence is called a **constant sequence**.

Example: Sequences

$$n \in \mathbb{N}, n \geq 1 \quad n \mapsto \frac{1}{n} \quad 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots$$

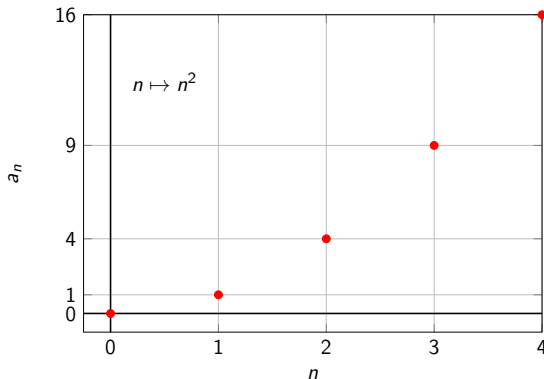
$$n \in \mathbb{N}, n \geq 2 \quad n \mapsto \frac{n+1}{n-1} \quad 3, 2, \frac{5}{3}, \frac{6}{4}, \frac{7}{5}, \dots$$

Definition of sequence and properties

We can represent sequences graphically by the **points** of the Cartesian plane with coordinates (n, a_n) .

Example: Graph of a sequence

The graph of the sequence $n \mapsto n^2$ with $n \in \{0, 1, 2, 3, 4\}$ is:



Definition of sequence and properties

To denote a sequence we will use the notation:

$$\{a_n\} \quad \text{or} \quad n \mapsto a_n$$

possibly specifying the set in which the input variable n varies (the whole set $\mathbb{N}$ or from a certain value $n_0 \in \mathbb{N}$ onward).

Definition of sequence and properties

A sequence $\{a_n\}$ is **bounded** if there exist two numbers $m \in \mathbb{R}$ and $M \in \mathbb{R}$ such that:

$$m \leq a_n \leq M, \quad \forall n \in \mathbb{N}$$

It is **bounded below** if m exists. It is **bounded above** if M exists.

Example: Bounded sequences

- ▶ the sequence $\{(-1)^n\}$ is bounded
- ▶ the sequence $\{n^2\}$ is only bounded below
- ▶ the sequence $\{(-2)^n\}$ is not bounded (neither below nor above).

Definition 2: Property that holds eventually

*We say that a sequence $\{a_n\}$ has (or acquires) a certain property **eventually** if there exists $\tilde{n} \in \mathbb{N}$ such that a_n satisfies that property for every $n \geq \tilde{n}$.*

Definition 3: Positive and negative sequences

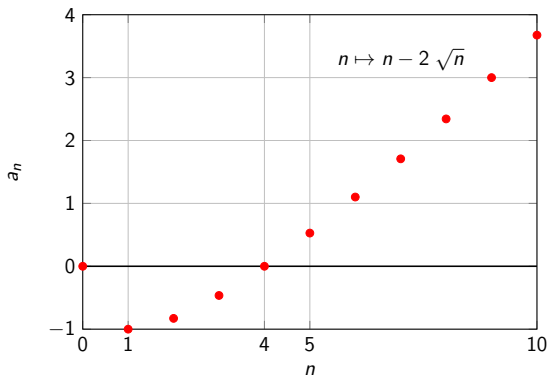
A sequence $\{a_n\}$ is called **non-negative** if $a_n \geq 0$ for every n ; **positive** if $a_n > 0$ for every n ; **non-positive** if $a_n \leq 0$ for every n ; **negative** if $a_n < 0$ for every n .

Definition 4: Eventually positive and negative sequences

A sequence $\{a_n\}$ is called **eventually non-negative** if there exists $\tilde{n} \in \mathbb{N}$ such that $a_n \geq 0$ for every $n \geq \tilde{n}$; **eventually positive** if there exists $\tilde{n} \in \mathbb{N}$ such that $a_n > 0$ for every $n \geq \tilde{n}$; **eventually non-positive** if there exists $\tilde{n} \in \mathbb{N}$ such that $a_n \leq 0$ for every $n \geq \tilde{n}$; **eventually negative** if there exists $\tilde{n} \in \mathbb{N}$ such that $a_n < 0$ for every $n \geq \tilde{n}$.

Example: Properties that hold eventually (1/2)

Consider the sequence $\{n - 2\sqrt{n}\}$. The graph of the sequence with $n \in \{0, 1, 2, \dots, 10\}$ is:

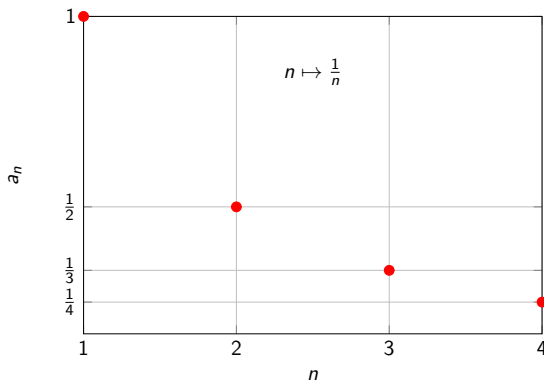


Example: Properties that hold eventually (2/2)

This sequence is eventually positive. With $n = 4$ we have $a_n = 0$, hence taking $\tilde{n} = 5$, we have $a_n > 0$ for $n \geq \tilde{n}$.

Example: Properties that hold eventually (1/2)

Now consider the sequence $\{\frac{1}{n}\}$. The graph of the sequence with $n \in \{1, 2, \dots, 4\}$ is:



Example: Properties that hold eventually (2/2)

This sequence is eventually less than 10^{-100} . With $n = 10^{100}$ we have $a_n = 10^{-100}$, hence taking for example $\tilde{n} = 10^{100} + 1$ we have $a_n < 10^{-100}$ for $n \geq \tilde{n}$.

Definition 5: Convergent sequence

A sequence $\{a_n\}$ is called **convergent** if there exists a number $\ell \in \mathbb{R}$ such that:

$$|a_n - \ell| < \varepsilon, \quad \text{eventually} \quad (1)$$

for every $\varepsilon > 0$.

Convergent sequences and definition of the limit of a sequence

A sequence $\{a_n\}$ is therefore called **convergent** if for every $\varepsilon > 0$ (as small as we like) there exists a number $n(\varepsilon) \in \mathbb{N}$ such that:

$$|a_n - \ell| < \varepsilon \quad \text{for every} \quad n \geq n(\varepsilon)$$

The number $n(\varepsilon)$ depends (in general) on the value of ε . If the sequence $\{a_n\}$ is convergent, then the number $\ell \in \mathbb{R}$ is associated with it.

Definition 6: Limit of a sequence

*The number $\ell \in \mathbb{R}$ appearing in inequality (1) is called the **limit of the sequence** $\{a_n\}$, and we write:*

$$\lim_{n \rightarrow +\infty} a_n = \ell \quad \text{or, equivalently} \quad a_n \rightarrow \ell \quad \text{for} \quad n \rightarrow +\infty$$

Convergent sequences and definition of the limit of a sequence

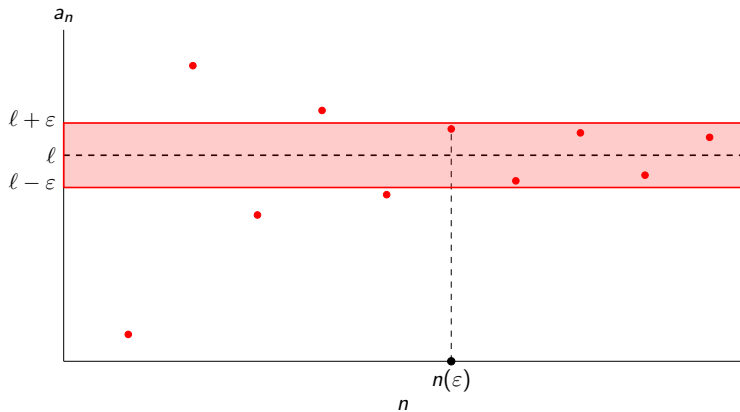
- read, respectively:

the limit of a_n , as n tends to infinity, is ℓ
 a_n tends to ℓ as n tends to infinity

Inequality (1) corresponds to the following two:

$$\ell - \varepsilon < a_n < \ell + \varepsilon \quad (2)$$

- Representing the points of a sequence graphically, we have:



The convergence condition means that, having fixed a horizontal strip “as narrow as we like”:

$$(\ell - \varepsilon, \ell + \varepsilon)$$

from a certain value of n onward, called $n(\varepsilon)$, the points a_n of the sequence no longer leave this strip. In the previous graph, having fixed the width of the strip, the values a_n lie inside the strip for $n \geq n(\varepsilon)$.

Theorem 1: Uniqueness of the limit of a sequence

If a sequence $\{a_n\}$ converges to the limit $\ell \in \mathbb{R}$, then this limit is unique.

Proof (1/2)

Suppose, by contradiction, that there exist two different limits, ℓ_1 and ℓ_2 , associated with the same sequence $\{a_n\}$. Then, eventually, for every $\varepsilon > 0$ we would have:

$$|\ell_1 - \ell_2| = |\ell_1 - a_n + a_n - \ell_2| \leq \underbrace{|\ell_1 - a_n|}_{=|a_n - \ell_1| < \varepsilon} + \underbrace{|a_n - \ell_2|}_{< \varepsilon} < 2\varepsilon \quad (3)$$

We used the triangle inequality. Since $\varepsilon > 0$ can be chosen arbitrarily small, (3) can be satisfied if and only if:

$$\ell_1 = \ell_2$$

Proof (2/2)

Hence two different values ℓ_1 and ℓ_2 cannot exist and consequently the limit (if it exists) is unique. □

Example: Verifying the limit of a sequence (1/4)

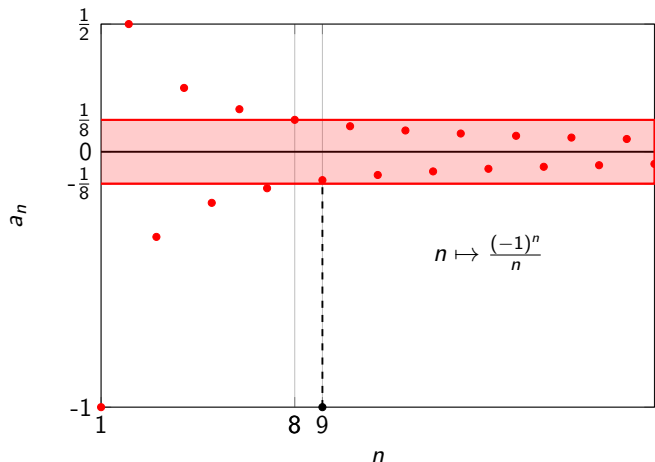
The graph of the sequence $n \mapsto \frac{(-1)^n}{n}$, for example starting from $n = 9$, lies within the horizontal strip:

$$\left(-\frac{1}{8}, \frac{1}{8}\right)$$

given by $\ell = 0$ and $\varepsilon = \frac{1}{8}$. With $n = 8$ we have $a_n = \frac{1}{8}$, with $n = 9$ we have $a_n = -\frac{1}{9}$, hence

$$|a_n| < \frac{1}{8} \quad \text{for every} \quad n \geq 9$$

Example: Verifying the limit of a sequence (2/4)



Example: Verifying the limit of a sequence (3/4)

To prove that:

$$a_n \rightarrow 0 \quad \text{for} \quad n \rightarrow +\infty$$

we must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n \geq n(\varepsilon) \Rightarrow |a_n| < \varepsilon$$

The inequality is equivalent to

$$\left| \frac{(-1)^n}{n} \right| < \varepsilon \quad \text{which is satisfied for} \quad n > \frac{1}{\varepsilon}$$

Example: Verifying the limit of a sequence (4/4)

Having fixed $\varepsilon > 0$, it will suffice to choose the first integer

$$n(\varepsilon) > \frac{1}{\varepsilon}$$

to satisfy the condition required by the definition of limit.

Observation 1

Every convergent sequence is bounded.

Proof (1/2)

Let $\{a_n\}$ be convergent to the limit $\ell \in \mathbb{R}$. Choosing $\varepsilon = 1$ in the definition of limit, there exists $n(1) \in \mathbb{N}$ such that

$$|a_n - \ell| < 1 \quad \text{for every} \quad n \geq n(1)$$

By the triangle inequality we then have

$$|a_n| = |(a_n - \ell) + \ell| \leq |a_n - \ell| + |\ell| < 1 + |\ell|, \quad n \geq n(1)$$

Proof (2/2)

The remaining terms $a_0, a_1, \dots, a_{n(1)-1}$ are *finitely many*, hence we can set

$$M = \max \{ |a_0|, \dots, |a_{n(1)-1}|, 1 + |\ell| \} \in \mathbb{R}$$

and we obtain

$$-M \leq a_n \leq M \quad \text{for every } n \in \mathbb{N}$$

that is, the sequence is bounded. □

Convergent sequences and definition of the limit of a sequence

- ▶ By contraposition: if a sequence is **not** bounded, then it is **not** convergent
- ▶ The converse does not hold: $\{(-1)^n\}$ is bounded but not convergent

Indeed, if for the sake of contradiction $(-1)^n \rightarrow \ell \in \mathbb{R}$, fixing $\varepsilon = \frac{1}{2}$ we should have $|(-1)^n - \ell| < \frac{1}{2}$ eventually, whereas

if $\ell \geq 0$, for every *odd* n : $|(-1)^n - \ell| = 1 + \ell \geq 1$

if $\ell < 0$, for every *even* n : $|(-1)^n - \ell| = 1 + |\ell| > 1$

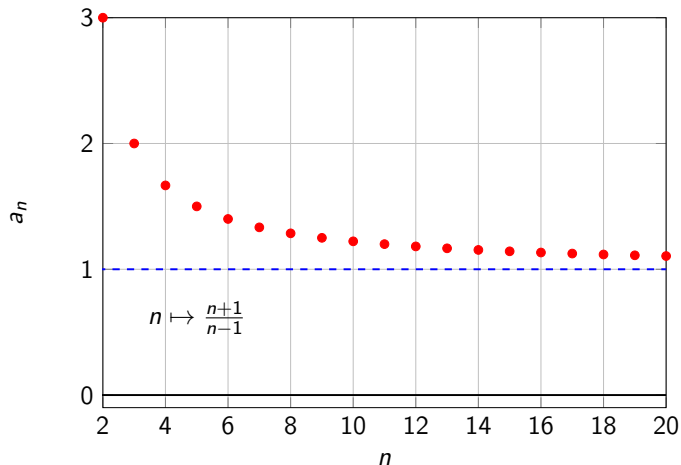
Example: Verifying the limit of a sequence (1/5)

Consider the sequence:

$$n \mapsto \frac{n+1}{n-1} \quad \left(\frac{n+1}{n-1} = \frac{n-1+1+1}{n-1} = 1 + \frac{2}{n-1} \right)$$

This is the graph of the sequence with $n \in \{2, 3, \dots, 20\}$:

Example: Verifying the limit of a sequence (2/5)



Example: Verifying the limit of a sequence (3/5)

We see that the values a_n get closer to 1, hence we try to prove, using the definition of limit, that:

$$\lim_{n \rightarrow +\infty} \frac{n+1}{n-1} = 1$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that:

$$n \geq n(\varepsilon) \Rightarrow 1 - \varepsilon < \frac{n+1}{n-1} < 1 + \varepsilon$$

Example: Verifying the limit of a sequence (4/5)

The left inequality is always satisfied (the numerator of the fraction is always larger than the denominator). We take the right one:

$$\frac{n+1}{n-1} - 1 < \varepsilon, \quad \frac{n+1 - (n-1)}{n-1} < \varepsilon, \quad \frac{2}{n-1} < \varepsilon, \quad n-1 > \frac{2}{\varepsilon}$$

$$\text{hence it is satisfied if} \quad n > \frac{2 + \varepsilon}{\varepsilon}$$

Example: Verifying the limit of a sequence (5/5)

Having fixed $\varepsilon > 0$, it will suffice to choose the first integer

$$n(\varepsilon) > \frac{2 + \varepsilon}{\varepsilon}$$

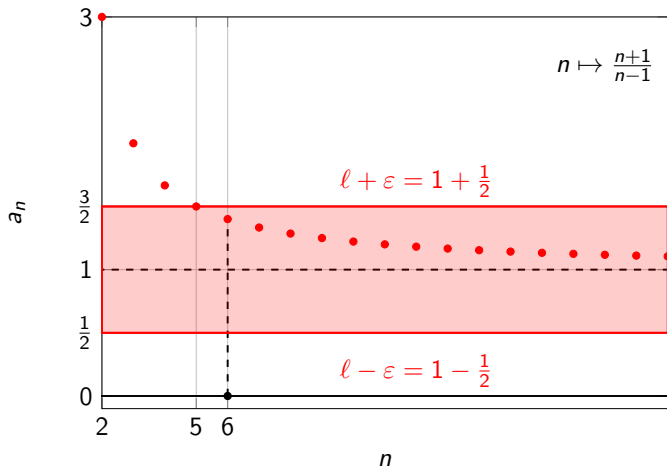
to satisfy the condition required by the definition of limit.

Example: Verifying the limit of a sequence (1/3)

For the previous example, we proved that the limit equals 1. Let us now check what happens when we fix $\varepsilon = \frac{1}{2}$. In this case

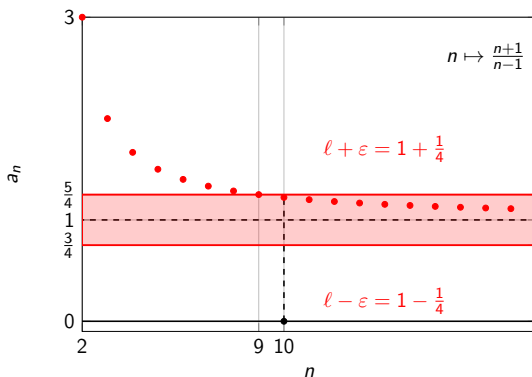
$$n\left(\frac{1}{2}\right) > \frac{2+1/2}{1/2} = 5.$$

Example: Verifying the limit of a sequence (2/3)



Example: Verifying the limit of a sequence (3/3)

If instead we fix $\varepsilon = \frac{1}{4}$, in this case we have $n\left(\frac{1}{4}\right) > \frac{2+1/4}{1/4} = 9$.



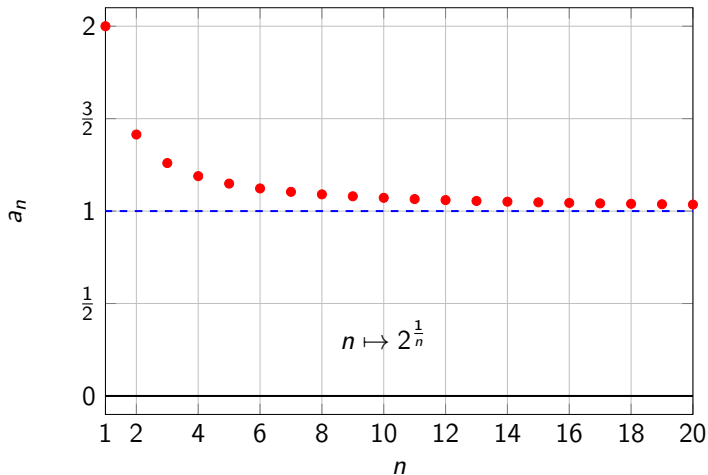
Example: Verifying the limit of a sequence (1/5)

Consider the sequence:

$$n \mapsto 2^{\frac{1}{n}}$$

This is the graph of the sequence with $n \in \{1, 2, \dots, 20\}$:

Example: Verifying the limit of a sequence (2/5)



Example: Verifying the limit of a sequence (3/5)

We see that the values a_n get closer to 1, hence we try to prove, using the definition of limit, that:

$$\lim_{n \rightarrow +\infty} 2^{\frac{1}{n}} = 1$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that:

$$n \geq n(\varepsilon) \Rightarrow 1 - \varepsilon < 2^{\frac{1}{n}} < 1 + \varepsilon.$$

Example: Verifying the limit of a sequence (4/5)

The left inequality is always satisfied (2 raised to a positive rational number), while for the right one, taking the logarithm to base 2, we obtain:

$$\frac{1}{n} < \log_2(1 + \varepsilon)$$

Hence it is satisfied if:

$$n > \frac{1}{\log_2(1 + \varepsilon)}$$

Example: Verifying the limit of a sequence (5/5)

Having fixed $\varepsilon > 0$, it suffices to choose the first integer

$$n(\varepsilon) > \frac{1}{\log_2(1 + \varepsilon)}$$

to satisfy the condition required by the definition of limit.

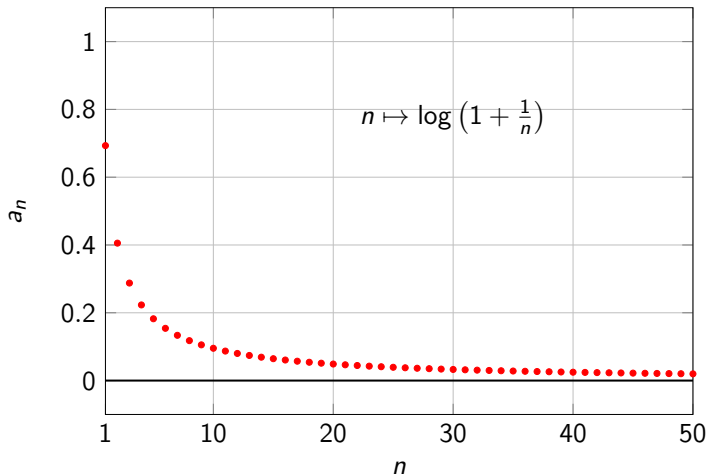
Example: Limit of sequences (1/5)

Consider the sequence:

$$n \mapsto \log \left(1 + \frac{1}{n} \right).$$

This is the graph of the sequence with $n \in \{1, 2, \dots, 50\}$:

Example: Limit of sequences (2/5)



Example: Limit of sequences (3/5)

We see that the values a_n get closer to 0, hence we try to prove, using the definition of limit, that:

$$\lim_{n \rightarrow +\infty} \log \left(1 + \frac{1}{n} \right) = 0$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that:

$$n \geq n(\varepsilon) \Rightarrow -\varepsilon < \log \left(1 + \frac{1}{n} \right) < \varepsilon$$

Example: Limit of sequences (4/5)

The left inequality is always satisfied (the logarithm to base e of a number greater than 1), while for the right one, exponentiating, we obtain:

$$\frac{1}{n} + 1 < e^\varepsilon$$

$$\text{hence it is satisfied if } n > \frac{1}{e^\varepsilon - 1}$$

$$(e^\varepsilon - 1 > 0 \text{ with } \varepsilon > 0)$$

Example: Limit of sequences (5/5)

Having fixed $\varepsilon > 0$, it suffices to choose the first integer

$$n(\varepsilon) > \frac{1}{e^\varepsilon - 1}$$

to satisfy the condition required by the definition of limit.

Definition 7: Sequence divergent to $+\infty$

A sequence $\{a_n\}$ is called **divergent** to $+\infty$ if for every $M > 0$ there exists a number $n(M) \in \mathbb{N}$ such that:

$$a_n > M \quad \text{for every } n \geq n(M)$$

Definition 8: Sequence divergent to $-\infty$

A sequence $\{a_n\}$ is called **divergent** to $-\infty$ if for every $M > 0$ there exists a number $n(M) \in \mathbb{N}$ such that:

$$a_n < -M \quad \text{for every } n \geq n(M)$$

Divergent sequences and irregular sequences

The number $n(M)$ depends (in general) on the value of M .

- In the two cases we will say, respectively, that $+\infty$ and $-\infty$ are the limits of the sequence and we will write, respectively:

$$\lim_{n \rightarrow +\infty} a_n = +\infty \quad \text{or} \quad \lim_{n \rightarrow +\infty} a_n = -\infty$$

The values of a sequence divergent to $+\infty$ eventually exceed any fixed real number. The values of a sequence divergent to $-\infty$ eventually fall below any fixed real number.

Divergent sequences and irregular sequences

The symbols $+\infty$ and $-\infty$ are not numbers.

- ▶ If we represent the real numbers on the **Euclidean line**, each number corresponds to a point and each point to a number.
- ▶ With the symbols $+\infty$ and $-\infty$ we agree to denote two “points”:
 - $+\infty$ lies to the *right* of every point of $\mathbb{R}$
 - $-\infty$ lies to the *left* of every point of $\mathbb{R}$however, no number corresponds to these two points.

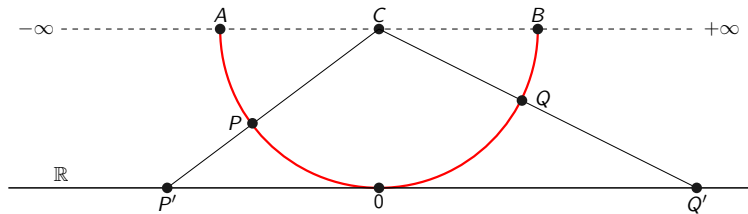
- ▶ On the symbols $+\infty$ and $-\infty$ the operations of sum and product with the properties stated in R_1 and R_2 are not defined, even though we will be able to perform these operations “partially” (as we will see later).

Definition 9: The set $\mathbb{R}^*$

The set of real numbers $\mathbb{R}$ with the addition of the two elements $+\infty$ and $-\infty$ will be denoted by:

$$\mathbb{R}^* = \mathbb{R} \cup \{+\infty\} \cup \{-\infty\}$$

- ▶ We can represent the set $\mathbb{R}^*$ “visually” by putting the points of the **line** in one-to-one correspondence with those of a **semicircle** (projecting them from the center of the semicircle onto the line $\mathbb{R}$):



Divergent sequences and irregular sequences

- ▶ No point on $\mathbb{R}$ corresponds to the points A and B ; we will say that $-\infty$ is the “counterpart” of the point A and $+\infty$ the “counterpart” of B .
- ▶ The limit operation becomes *fully meaningful* if set in $\mathbb{R}^*$ instead of $\mathbb{R}$, i.e., the limit of a sequence can be a real number ℓ or $+\infty$ or $-\infty$.

Sequences whose limit is a real number are **convergent**; those whose limit is $+\infty$ or $-\infty$ are **divergent**.

Example: Convergent and divergent sequences

- ▶ the canonical sequence $\{n\}$ is divergent to $+\infty$;
- ▶ the sequence $\{2^n\}$ is divergent to $+\infty$;
- ▶ the sequence $\{-2^n\}$ is divergent to $-\infty$;
- ▶ the sequence $\{2^{\frac{1}{n}}\}$ is convergent to 1.

Observation 2

$$\lim_{n \rightarrow +\infty} n^{\alpha} = \begin{cases} +\infty & \text{if } \alpha > 0 \\ 1 & \text{if } \alpha = 0 \\ 0 & \text{if } \alpha < 0 \end{cases}$$

Proof (1/4)

If $\alpha > 0$, we must prove that the sequence diverges to $+\infty$. We must therefore verify that for every $M > 0$ there exists $n(M) \in \mathbb{N}$ such that

$$n \geq n(M) \Rightarrow a_n > M$$

The inequality

$$n^\alpha > M \quad \text{is satisfied for } n > M^{\frac{1}{\alpha}}$$

Proof (2/4)

Hence, having fixed $M > 0$, it will suffice to choose the first integer

$$n(M) > M^{\frac{1}{\alpha}}$$

to satisfy the required divergence condition.

Hence there is no $M > 0$ such that $n^\alpha \leq M$ for $n \in \mathbb{N}$, and the sequence is not bounded. Hence $n^\alpha \rightarrow +\infty$ for $n \rightarrow +\infty$.

Proof (3/4)

If $\alpha < 0$, we must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n \geq n(\varepsilon) \Rightarrow -\varepsilon < n^\alpha < \varepsilon$$

The left inequality is always satisfied. For the right inequality, since $n^\alpha = \frac{1}{n^{|\alpha|}}$, we obtain

$$\frac{1}{n^{|\alpha|}} < \varepsilon$$

Proof (4/4)

which is satisfied if

$$n^{|\alpha|} > \frac{1}{\varepsilon} \quad \text{i.e., for } n > \frac{1}{\varepsilon^{1/|\alpha|}}$$

Having fixed $\varepsilon > 0$, it will suffice to choose the first integer

$$n(\varepsilon) > \frac{1}{\varepsilon^{1/|\alpha|}}$$

to satisfy the condition required by the definition of limit. □

- Finally, we observe that there are sequences that are neither convergent nor divergent

Definition 10: Irregular (indeterminate) sequence

*A sequence that is neither convergent nor divergent is called **irregular** (oscillating) or **indeterminate**.*

Example: Irregular sequences

- ▶ the sequence $\{(-1)^n\}$ is neither convergent nor divergent (but it is bounded)
- ▶ the sequence $\{(-2)^n\}$ is neither convergent nor divergent (and not even bounded).

Proof

We check that $\{(-1)^n\}$ is irregular.

It is not convergent: we saw it above, as a counterexample to the converse of “convergent $\Rightarrow$ bounded”

It is not divergent: for every $n \in \mathbb{N}$ we have $-1 \leq (-1)^n \leq 1$. Fixing $M = 1$, for every *odd* n we get $(-1)^n = -1 < M$, hence $(-1)^n \rightarrow +\infty$ is impossible; for every *even* n we get $(-1)^n = 1 > -M$, hence $(-1)^n \rightarrow -\infty$ is impossible

The limit therefore does not exist. □

Divergent sequences and irregular sequences

- For **irregular** sequences, the limit operation is not defined, i.e., their **limit does not exist**.

Summarizing, the operation of computing the limit allows us to answer rigorously the question: **how do the numbers a_n behave as n becomes larger and larger?**

Summary (1/2)

Sequences:

- ▶ A sequence is a function $f : \mathbb{N} \rightarrow \mathbb{R}$, $n \mapsto a_n$ (possibly from some n_0 onwards)
- ▶ It is represented by the points (n, a_n) of the Cartesian plane
- ▶ It is **bounded** if $m \leq a_n \leq M$ for all n ; a property holds **eventually** if it holds for all $n \geq \tilde{n}$

Convergent sequences:

- ▶ $a_n \rightarrow \ell$ if $\forall \varepsilon > 0$ we have $|a_n - \ell| < \varepsilon$ eventually
- ▶ The limit, if it exists, is **unique**; convergent sequences are bounded

Summary (2/2)

Divergent and irregular sequences:

- ▶ $a_n \rightarrow +\infty$ if $\forall M > 0$ we have $a_n > M$ eventually (similarly for $-\infty$)
- ▶ In the set $\mathbb{R}^* = \mathbb{R} \cup \{+\infty\} \cup \{-\infty\}$ the limit of a sequence can be a real number or $\pm\infty$
- ▶ $\lim n^\alpha$ equals $+\infty$, 1 or 0 according to whether $\alpha > 0$, $\alpha = 0$ or $\alpha < 0$
- ▶ Sequences that neither converge nor diverge are **irregular** (for example $(-1)^n$)

Unbounded sets and their suprema/infima

Introduction: Unbounded sets and their suprema/infima

- ▶ In this section we extend the notions of sup and inf to **unbounded** sets
- ▶ Using the symbols $\pm\infty$ of $\mathbb{R}^*$, every non-empty subset of $\mathbb{R}$ will have a supremum and an infimum
- ▶ This convention simplifies the statements of the theorems on limits, in particular the one on monotone sequences

- It is convenient to adopt the convention introduced for limits also for the sup and the inf, extending the definition of these quantities as follows

Definition 11: Supremum and infimum sup and inf (unbounded sets)

If a set $E \subseteq \mathbb{R}$ is not bounded above (below), we will say that

$$\sup E = +\infty \quad (\inf E = -\infty)$$

Unbounded sets and their suprema/infima

- In this way, property R_4 of the real numbers can be stated as follows:

$R_4 \rightarrow$ every non-empty set $E \subseteq \mathbb{R}$ has a supremum and an infimum; $\sup E$ ($\inf E$) is a number if E is bounded above (below), otherwise it is $+\infty$ ($-\infty$).

Suprema and infima of unbounded sets:

- ▶ If $E \subseteq \mathbb{R}$ is not bounded above we set $\sup E = +\infty$
- ▶ If E is not bounded below we set $\inf E = -\infty$
- ▶ Hence every non-empty set $E \subseteq \mathbb{R}$ has a supremum and an infimum in $\mathbb{R}^*$

Infinitesimal and infinite sequences



Introduction: Infinitesimal and infinite sequences

- ▶ In this section we give a name to sequences tending to 0 (**infinitesimal**) and to those tending to $\pm\infty$ (**infinite**)
- ▶ We also specify whether a sequence approaches its limit **from above** or **from below**
- ▶ These concepts are the basis for comparing infinitesimals and infinities, which we will use to compute limits

Definition 12: Infinitesimal sequence

*A sequence $\{a_n\}$ tending to zero is called **infinitesimal***

Example: Infinitesimal sequences

- ▶ the sequence $\left\{\frac{1}{n}\right\}$ is infinitesimal
- ▶ the sequence $\left\{\frac{1}{n^2}\right\}$ is infinitesimal

Theorem 2: On infinitesimal sequences

A sequence $\{a_n\}$ is infinitesimal if and only if the sequence of absolute values $\{|a_n|\}$ is infinitesimal:

$$\lim_{n \rightarrow +\infty} a_n = 0 \iff \lim_{n \rightarrow +\infty} |a_n| = 0.$$

Proof

If $\{a_n\}$ is infinitesimal, for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$\underbrace{|a_n - 0|} < \varepsilon, \quad n \geq n(\varepsilon)$$
$$= \left| |a_n| - 0 \right|$$

which is exactly the condition $|a_n| \rightarrow 0$. Conversely, if $|a_n| \rightarrow 0$,

$$\underbrace{\left| |a_n| - 0 \right|} < \varepsilon, \quad n \geq n(\varepsilon)$$
$$= |a_n - 0|$$

which is exactly the condition $a_n \rightarrow 0$.



Infinitesimal and infinite sequences

- ▶ Beware: the theorem holds **only** for the limit 0
- ▶ For instance $|(-1)^n| = 1 \rightarrow 1$, but $\{(-1)^n\}$ is not convergent

Infinitesimal and infinite sequences

- ▶ The concept of *infinitesimal* plays a central role and is also fundamental for having a correct and effective **intuitive picture** of the concepts of *infinitesimal calculus*.

an “**infinitesimal**” is not an “infinitely small number” (a meaningless concept) but a **variable quantity** (a sequence or, as we will see, a function) that **becomes indefinitely small**.

Definition 13: Infinite sequence

A sequence $\{a_n\}$ tending to $\pm\infty$ is called *infinite* (an infinity)

Example: Infinite sequences

- ▶ the sequence $\{n^2\}$ is infinite
- ▶ the sequence $\{n!\}$ is infinite

- ▶ Sometimes it is possible to specify whether a convergent sequence **approaches its limit from above or from below**

Definition 14: Limit from above

We say that the sequence $\{a_n\}$ tends to $\ell \in \mathbb{R}$ **from above** and we write

$$\lim_{n \rightarrow +\infty} a_n = \ell^+$$

if for every $\varepsilon > 0$ we have that

$$0 \leq a_n - \ell < \varepsilon, \quad \text{eventually.}$$

Definition 15: Limit from below

We say that the sequence $\{a_n\}$ tends to $\ell \in \mathbb{R}$ **from below** and we write

$$\lim_{n \rightarrow +\infty} a_n = \ell^-$$

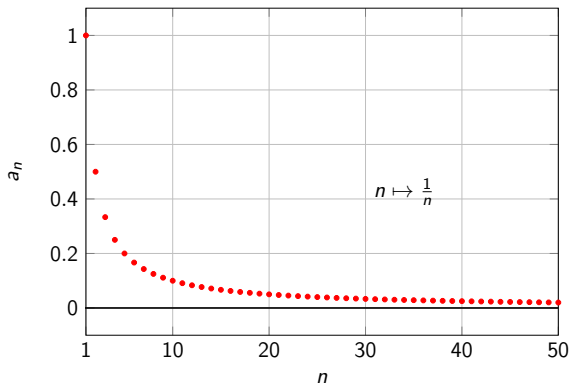
if for every $\varepsilon > 0$ we have that

$$0 \leq \ell - a_n < \varepsilon, \quad \text{eventually.}$$

- ▶ Saying that $a_n \rightarrow \ell^+$ for $n \rightarrow +\infty$ means stating that $a_n \rightarrow \ell$ and moreover $a_n \geq \ell$ eventually; hence a_n approaches ℓ *from above*, i.e., it approximates ℓ from above.
- ▶ Saying that $a_n \rightarrow \ell^-$ for $n \rightarrow +\infty$ means stating that $a_n \rightarrow \ell$ and moreover $a_n \leq \ell$ eventually; hence a_n approaches ℓ *from below*, i.e., it approximates ℓ from below.

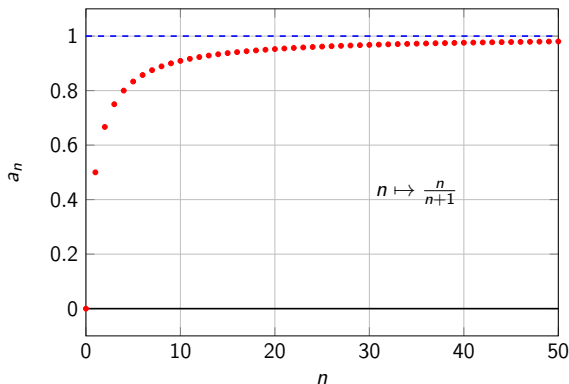
Example: Sequence with limit from above

$$\lim_{n \rightarrow +\infty} \frac{1}{n} = 0^+$$



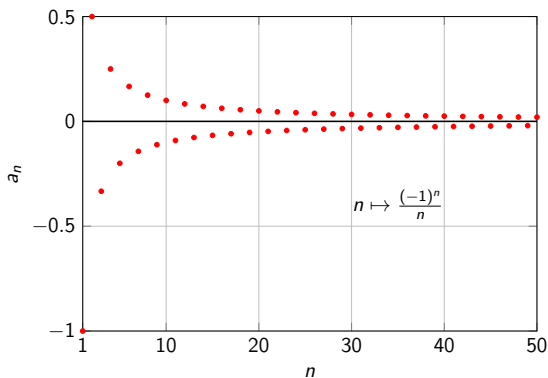
Example: Sequence with limit from below

$$\lim_{n \rightarrow +\infty} \frac{n}{n+1} = 1^- \quad \left(\frac{n}{n+1} = \frac{n+1-1}{n+1} = 1 - \frac{1}{n+1} \right)$$



Example: Sequence with a limit, but neither from below nor from above (1/2)

$$\lim_{n \rightarrow +\infty} \frac{(-1)^n}{n} = 0$$



Example: Sequence with a limit, but neither from below nor from above (2/2)

In this case we can state neither that $a_n \rightarrow 0^+$ nor that $a_n \rightarrow 0^-$ for $n \rightarrow +\infty$.

Summary

Infinitesimals and infinities:

- ▶ $\{a_n\}$ is **infinitesimal** if $a_n \rightarrow 0$, **infinite** if $a_n \rightarrow \pm\infty$
- ▶ An infinitesimal is a **variable quantity** that becomes as small as we like

Limit from above and from below:

- ▶ $a_n \rightarrow \ell^+$ if $0 \leq a_n - \ell < \varepsilon$ eventually, $a_n \rightarrow \ell^-$ if $0 \leq \ell - a_n < \varepsilon$ eventually
- ▶ Examples: $\frac{1}{n} \rightarrow 0^+$, $\frac{n}{n+1} \rightarrow 1^-$, while $\frac{(-1)^n}{n} \rightarrow 0$ neither from above nor from below

Monotone sequences



Introduction: Monotone sequences

- ▶ In this section we study **monotonic** sequences (non-decreasing or non-increasing)
- ▶ We prove the **monotone sequence theorem**: a monotone sequence always has a limit, which is finite if the sequence is bounded
- ▶ It is a fundamental tool to prove that a limit exists even without knowing its value (we will use it to define the number e)

Definition 16: Monotone sequences

A sequence $\{a_n\}$ is called **non-decreasing** if:

$$a_n \leq a_{n+1}, \forall n$$

A sequence $\{a_n\}$ is called **increasing** if:

$$a_n < a_{n+1}, \forall n$$

A sequence $\{a_n\}$ is called **non-increasing** if:

$$a_n \geq a_{n+1}, \forall n$$

A sequence $\{a_n\}$ is called **decreasing** if:

$$a_n > a_{n+1}, \forall n$$

Monotone sequences

- ▶ A sequence is **monotonic** if it is non-decreasing or non-increasing
- ▶ It is **strictly monotonic** if it is increasing or decreasing

Definition 17: Eventually monotone sequences

There exists $\tilde{n} \in \mathbb{N}$ such that, for every $n \geq \tilde{n}$:

$$a_n \leq a_{n+1} \quad (\text{eventually non-decreasing})$$

$$a_n < a_{n+1} \quad (\text{eventually increasing})$$

$$a_n \geq a_{n+1} \quad (\text{eventually non-increasing})$$

$$a_n > a_{n+1} \quad (\text{eventually decreasing})$$

Example: Increasing and decreasing sequences

- ▶ The sequence $\{n^2\}$ is increasing
- ▶ The sequence $\{\frac{1}{n}\}$ is decreasing
- ▶ The sequence $\{(-1)^n\}$ is not monotonic
- ▶ every constant sequence is monotonic (it is both non-decreasing and non-increasing), but not strictly monotonic

- ▶ With regard to the limit operation, these sequences are of particular importance; indeed, they are **never irregular**, but are **convergent or divergent** depending on whether they are **bounded or not**.

Theorem 3: Monotone sequence theorem

- ▶ Let $\{a_n\}$ be a **non-decreasing sequence that is bounded above**. Then $\{a_n\}$ is convergent, and its limit equals

$$\sup\{a_n : n \in \mathbb{N}\}.$$

- ▶ Let $\{a_n\}$ be a **non-increasing sequence that is bounded below**. Then $\{a_n\}$ is convergent, and its limit equals

$$\inf\{a_n : n \in \mathbb{N}\}.$$

Proof (1/5)

We consider the case of non-decreasing sequences that are bounded above.

Since the sequence is bounded above, the set of values taken by the sequence $\{a_n : n \in \mathbb{N}\}$ is bounded above.

By the *supremum property* enjoyed by $\mathbb{R}$, there therefore exists a finite supremum, which we denote by ℓ :

$$\ell = \sup\{a_n : n \in \mathbb{N}\} \quad \text{and} \quad \ell \in \mathbb{R}.$$

Proof (2/5)

We now need to prove that

$$\lim_{n \rightarrow +\infty} a_n = \ell$$

and hence that for every $\varepsilon > 0$ we have

$$\ell - \varepsilon < a_n < \ell + \varepsilon, \quad \text{eventually.}$$

The second inequality is obvious. For every $n \in \mathbb{N}$ we have

$$a_n \leq \ell \quad (\text{and hence } a_n < \ell + \varepsilon)$$

Proof (3/5)

since ℓ is the supremum of $\{a_n : n \in \mathbb{N}\}$ and hence an upper bound. By definition of supremum, ℓ is the least of the upper bounds of the set $\{a_n : n \in \mathbb{N}\}$. Therefore, since

$$\ell - \varepsilon < \ell,$$

certainly $\ell - \varepsilon$ is not an upper bound of the set $\{a_n : n \in \mathbb{N}\}$. This means that there exists an $n(\varepsilon) \in \mathbb{N}$ for which

$$a_{n(\varepsilon)} > \ell - \varepsilon.$$

Proof (4/5)

On the other hand, the sequence is non-decreasing, therefore for every $n \geq n(\varepsilon)$ we have $a_n \geq a_{n(\varepsilon)}$. We have thus proved that

$$a_n \geq a_{n(\varepsilon)} > \ell - \varepsilon \quad \text{for every } n \geq n(\varepsilon)$$

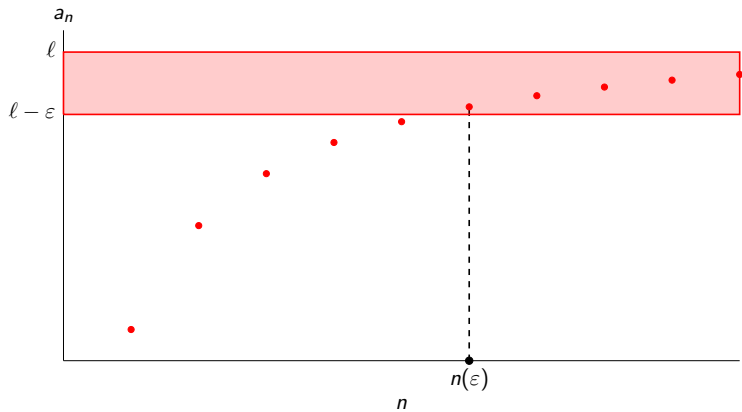
Consequently, we have the claim:

$$\lim_{n \rightarrow +\infty} a_n = \ell$$

Proof (5/5)

The case of non-increasing sequences that are bounded below is proved analogously. □

- ▶ The idea of the proof is conveyed by the following figure:



- To express symbolically as well that the limit is the sup (or the inf) of a non-decreasing (or non-increasing) sequence, we use the notation

$$a_n \uparrow \ell \quad \text{or} \quad a_n \downarrow \ell$$

This in particular implies that $a_n \rightarrow \ell^-$ (respectively, ℓ^+), but it contains a further piece of information: the monotonicity of the sequence.

- ▶ This theorem is a consequence of the completeness axiom (axiom of continuity) R_4 of the real numbers and therefore holds if the setting we consider is $\mathbb{R}$. For example, it is not true that an increasing and bounded sequence of rational numbers always has a rational limit, i.e., in $\mathbb{Q}$.

Example: Increasing and bounded sequence in $\mathbb{Q}$ ($1/2$)

Let $\{a_n\}$ be the sequence defined as follows:

$$a_0 = 0, a_1 = 0, 1, a_2 = 0, 1011, a_3 = 0, 10110111, a_4 = 0, 1011011101111 \dots$$

At step n we append to the decimal number obtained at the previous step a digit zero followed by n digits equal to 1. The sequence $\{a_n\}$ is evidently increasing, and bounded above (for example, $a_n \leq 1$).

Example: Increasing and bounded sequence in $\mathbb{Q}$ (2/2)

- ▶ In $\mathbb{R}$ the sequence converges to the number $\sup\{a_n : n \in \mathbb{N}\}$, which after the decimal point has an infinite, non-periodic decimal expansion (one digit 1, one digit 0, two digits 1, one digit 0, three digits 1, one digit 0, and so on forever)
- ▶ Hence the limit of the sequence is an irrational number. This example shows that in the set $\mathbb{Q}$ the monotone sequence theorem is false.

- ▶ The monotone sequence theorem can be completed with the next corollary, which considers bounded or unbounded sequences.

Corollary 1: Of the monotone sequence theorem

Let $\{a_n\}$ be a non-decreasing sequence. Then there exists

$$\lim_{n \rightarrow +\infty} a_n = \sup\{a_n : n \in \mathbb{N}\}.$$

Proof (1/2)

If $\{a_n\}$ is bounded above, the statement is contained in Theorem (monotone sequence theorem).

If instead $\{a_n\}$ is unbounded above, this means that, having fixed $M > 0$, there exists an $n(M) \in \mathbb{N}$ such that

$$a_{n(M)} > M$$

On the other hand, the sequence is non-decreasing, therefore for every $n \geq n(M)$ we have

$$a_n \geq a_{n(M)} > M$$

Proof (2/2)

We have thus proved that for every $M > 0$ we have $a_n > M$, eventually. This means that $a_n \rightarrow +\infty$ for $n \rightarrow +\infty$. □

Monotone sequences

Summarizing, we have:

- ▶ if $\{a_n\}$ is bounded above, then it converges (and its limit equals the supremum of its values, which in this case is a real number)
- ▶ if instead $\{a_n\}$ is unbounded above, then a_n tends to $+\infty$ (which in this case equals the supremum of its values).

A monotonic sequence either converges or diverges (it cannot be irregular).

Summary

Monotone sequences:

- ▶ Non-decreasing if $a_n \leq a_{n+1}$, non-increasing if $a_n \geq a_{n+1}$
(increasing and decreasing with $<$, $>$)
- ▶ **Monotone sequence theorem:** if non-decreasing and bounded above $a_n \rightarrow \sup\{a_n\}$; if non-increasing and bounded below $a_n \rightarrow \inf\{a_n\}$
- ▶ The theorem relies on the completeness axiom of $\mathbb{R}$: it is false in $\mathbb{Q}$
- ▶ **Corollary:** a non-decreasing sequence always has limit $\sup\{a_n\} \in \mathbb{R}^*$
- ▶ A monotonic sequence converges or diverges, it is never irregular

Limits of geometric progressions

Introduction: Limits of geometric progressions

- ▶ In this section we study the **geometric progression** $\{a^n\}$ with ratio $a \in \mathbb{R}$
- ▶ We compute its limit as a varies, verifying it with the definition
- ▶ It is one of the fundamental limits: we will meet it again in the ratio test and in geometric series

- ▶ Consider the **geometric progression** with common ratio $a \in \mathbb{R}$:

$$1, a, a^2, a^3, \dots, a^n, \dots$$

it is equivalent to the sequence $\{a^n\}$:

$$n \mapsto a^n$$

- ▶ If $a > 1$, the sequence is increasing and unbounded above.
- ▶ If $a = 1$, the sequence is constant.

- ▶ If $0 < a < 1$, the sequence is decreasing and tends to zero.
- ▶ If a is negative, the sequence is not monotonic.

Observation 3

$$\lim_{n \rightarrow +\infty} a^n = \begin{cases} +\infty & \text{if } a > 1 \\ 1 & \text{if } a = 1 \\ 0 & \text{if } |a| < 1 \\ \text{does not exist} & \text{if } a \leq -1 \end{cases}$$

Proof (1/6)

If $a = 0$, we have $a^n = 0$ for every $n \geq 1$: the sequence is eventually zero and therefore $a^n \rightarrow 0$.

If $0 < |a| < 1$, we must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n \geq n(\varepsilon) \Rightarrow |a^n| < \varepsilon$$

Since $|a^n| = |a|^n$, we must verify that

$$n \geq n(\varepsilon) \Rightarrow |a|^n < \varepsilon$$

Proof (2/6)

Taking the logarithm in base $|a|$, which is *decreasing* since $0 < |a| < 1$, the inequality is satisfied if

$$n > \log_{|a|} \varepsilon$$

Having fixed $\varepsilon > 0$, it will suffice to choose the first integer

$$n(\varepsilon) > \log_{|a|} \varepsilon$$

to satisfy the condition required by the definition of limit.

Proof (3/6)

If $a > 1$, we must verify that for every $M > 0$ there exists $n(M) \in \mathbb{N}$ such that

$$n \geq n(M) \Rightarrow a_n > M$$

The inequality

$$a^n > M \quad \text{is satisfied for } n > \log_a M$$

Proof (4/6)

Hence, having fixed $M > 0$, it will suffice to choose the first integer

$$n(M) > \log_a M$$

to satisfy the required divergence condition.

Proof (5/6)

If $a \leq -1$, we have $|a| \geq 1$ and therefore

$$a^n = (-1)^n |a|^n \quad \text{with} \quad |a|^n \geq 1, \quad \forall n \in \mathbb{N}$$

that is, $a^n \geq 1$ for n even and $a^n \leq -1$ for n odd.

It is not convergent: if, for the sake of contradiction, $a^n \rightarrow \ell \in \mathbb{R}$, then fixing $\varepsilon = \frac{1}{2}$ we should have $|a^n - \ell| < \frac{1}{2}$ eventually. Instead, if $\ell \geq 0$, for every *odd* n , $|a^n - \ell| = |a|^n + \ell \geq 1$; if $\ell < 0$, for every *even* n , $|a^n - \ell| = |a|^n + |\ell| > 1$.

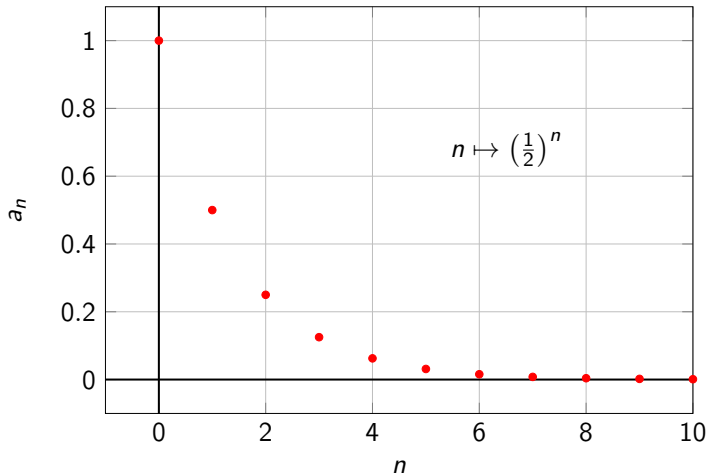
Proof (6/6)

It is not divergent: fixing $M = 1$, for every *odd* n we have $a^n \leq -1 < M$, hence $a^n \rightarrow +\infty$ is impossible; for every *even* n we have $a^n \geq 1 > -M$, hence $a^n \rightarrow -\infty$ is impossible.

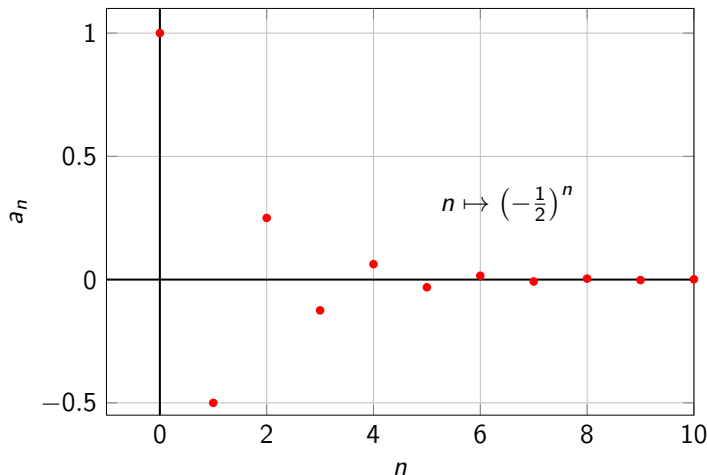
The limit therefore does not exist.



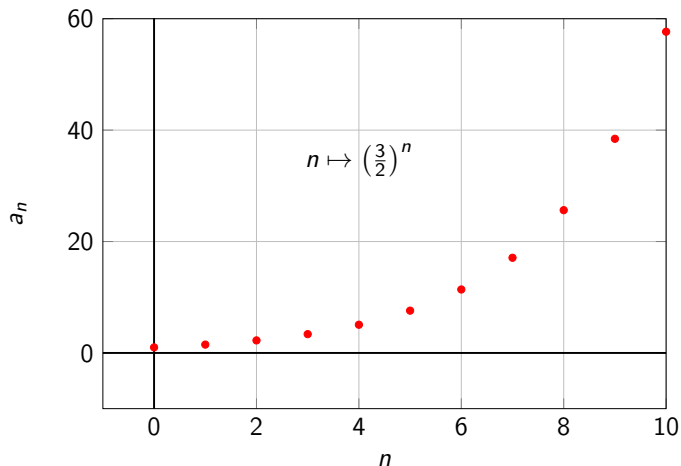
Example: Infinitesimal and decreasing geometric progression



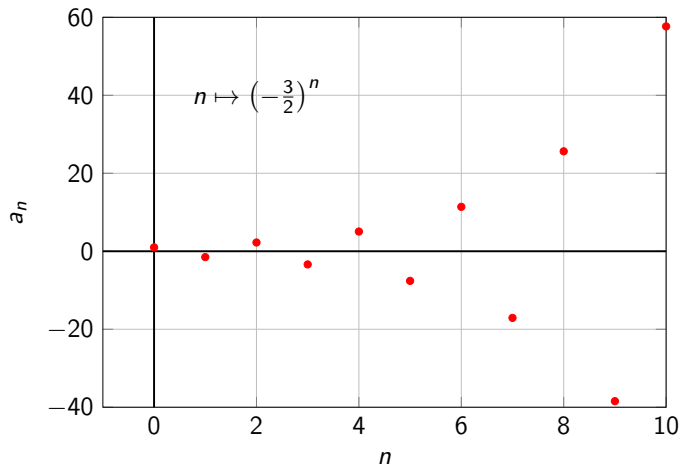
Example: Infinitesimal (but not monotonic) geometric progression



Example: Divergent geometric progression



Example: Geometric progression that is neither convergent nor divergent



Geometric progression $\{a^n\}$:

- ▶ $a^n \rightarrow +\infty$ if $a > 1$
- ▶ $a^n \rightarrow 1$ if $a = 1$ (constant sequence)
- ▶ $a^n \rightarrow 0$ if $|a| < 1$ (monotonic if $0 < a < 1$, with alternating signs if $-1 < a < 0$)
- ▶ The limit does not exist if $a \leq -1$

Summary (1/2)

Sequences and limits:

- ▶ A sequence is a function $n \mapsto a_n$ defined on the natural numbers
- ▶ Convergent ($a_n \rightarrow \ell \in \mathbb{R}$), divergent ($a_n \rightarrow \pm\infty$) or irregular
- ▶ The limit, if it exists, is unique

Suprema and infinitesimals:

- ▶ $\sup E = +\infty$ and $\inf E = -\infty$ for unbounded sets
- ▶ Infinitesimals ($a_n \rightarrow 0$), infinities ($a_n \rightarrow \pm\infty$), limits from above and from below

Summary (2/2)

Monotone sequences:

- ▶ They always have a limit: $\sup\{a_n\}$ if non-decreasing, $\inf\{a_n\}$ if non-increasing

Geometric progressions:

- ▶ $a^n \rightarrow +\infty$ ($a > 1$), 1 ($a = 1$), 0 ($|a| < 1$), no limit ($a \leq -1$)